CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Fifth Semester of B. Tech. (EC) Examination

November 2013

EC302 Integrated Circuits & Applications

Date: 20.11.2013, Wednesday **Time: 10:00 a.m. To 01:0 0 p.m.** **Maximum Marks: 70**

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION – I

Q - 1 (a) Why open loop op-amp configuration is not suitable for linear applications? [03]

(b) Define Slew rate and explain the significance of its with reference to op-amp. [02]

(c) Draw the circuit diagram of current series negative feedback amlifier. [02]

Q - 2 (a) (i) Design an op-amp differentiator that will differentiate an input signal with f_{max}= 1 kHz. [07]

(ii) Draw the output waveform for a sine wave of 1V peak at 1 kHz applied to the differentiator.

(iii) Repeat part (ii) for a square wave input.

(b) Expalin the block diagram of typical op-amp. [04]

(c) The 741 op-amp having the following parameters is connected as a inverting amplifier with R_1=10 kΩ and R_F=100kΩ: [03]

The input offset voltage =12mV, The biasing current = 500nA and the input offset current =90nA.

Evaluate the values of total output offset voltage, compensating resistance required to compensate effect of input bias current, total output offset voltage with compensating resistor.

OR

Q - 2 (a) What is difference between clippers and clampers? Explain positive and negative clipper circuit in details with input and output waveforms. [06]

(b) Design a scaling circuit to scale 10% of input V_1,20% of input V_2 and 70% of input V_3 at output terminal of op-amp. [05]

(c) Explain two special cases of voltage shunt feedback amplifier using op-amp. [03]

Q-3 (a) What is difference between the sawtooth wave and the triangular wave? [07]

For triangular wave generator prove that $f_o=R_3/4R_1C_1R_2$.

(b) In Schmitt trigger circuit, calculate the values of R_1 and R_2 if saturation voltages are +12V and -12V. Assume hysteresis width=6 V. [04]

(c) What determines the peak frequency f_p in the peaking amplifier? [03]

OR

Q - 3 (a) What do you understand by precision rectifiers? Draw and explain half wave precision rectifier circuit using op-amp. [06]

(b) Design an astable multivibrator having an o/p frequency of 10khz with a duty cycle of 25%. [05]

(b) List the important characteristics of the comparator. [03]

SECTION – II

Q - 4 (a) What are the advantages of active filters over passive ones? [02]

(b) Explain performance parameters with reference to voltage regulator. [03]

(c) Draw and explain the magnitude and attenuation characteristics of band pass filter. [02]

Q - 5 (a) Specifications for Butter worth low pass filter are , α_{max}= 0.2 dB, α_{min}= 20 dB, ω_p= 8000 rad/sec, w_s=16,000 rad/sec. [07]

Determine n, half power frequency (ω_0) and actual attenuation at the edge of pass band $\alpha(\omega_p)$ and stop band $\alpha(\omega_s)$.

(b) Using the LM317, design an adjustable voltage regulator to satisfy the following specifications: V_0 = 10.5V to 18.75V, output current I_0=0.5A, I_{ADJ}=100μA ,R_1=240Ω and V_{REF}=1.25V. Draw the complete schematic diagram of the designed regulator. [04]

(c) Using the 7805C voltage regulator, design a current source that will deliver 130mA current to the 10Ω, load. Draw the complete schematic diagram of the circuit. [03]

OR

Q - 5 (a) What are the basic building blocks of a PLL? Explain the need for each component in details. [07]

(b) Design a circuit to realize the transfer function of the Bode plot given in Fig. (1). Use inverting configuration. [07]

Q - 6 (a) Derive frequency response of Sallen & Key circuit and from that identify the type of filter. Draw the circuit when C_1=C_2=1 and R_1=R_2=1. [07]

(b) Explain application of 555 timer IC as 'a divide by 2' frequency divider. And mention clearly the relationship between the pulse width t_p and the period T of the input trigger signal. [07]

OR

Q - 6 (a) Design Tow Thomas Biquad circuit. Is it orthogonally tuned circuit? Justify your answer. [07]

(b) List one application of PLL and then briefly describe the role of the PLL in that application. [07]

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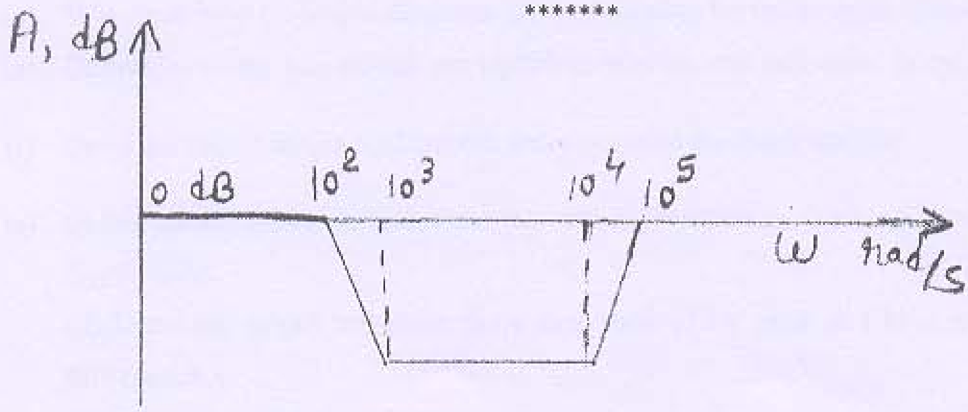

Fig. (1)